BOOSTER PUMPS: FAULT FINDING AND ALARMS

The alarms a booster pump set raises, what each one usually means, and what the technician checks first. Work through the checks in the order given.

High motor temperature trip

Check first whether the drive cabinet fan is running and the cabinet filters are clear. A blocked filter is the cause in most cases, and the trip clears after the cabinet has cooled. Then check that the pump is not running against a closed discharge valve, which heats the water in the casing. Only then suspect the motor itself; a motor fault is referred to the maintenance engineer.

Low suction pressure trip

The upstream reservoir is low or an upstream valve is closed. Check the reservoir level on the station display before anything else. Do not override the trip: running a booster pump dry destroys the mechanical seal within minutes.

High discharge pressure alarm

Usually a closed valve downstream or a burst that has been isolated. Check the network diagram and the recent work orders for the zone. The drive should have reduced speed; if it did not, the pressure transmitter is suspect.

Drive fault codes

Fault codes are listed in the drive manufacturer's manual carried in each station. Record the code before resetting; a code that recurs three times in a shift is a repair work order, not a reset.

Seal leak

A steady drip from the seal housing is normal for a new seal for the first hours. A stream is

a seal failure: stop the pump, switch duty, and raise a repair work order. The seal replacement and the pressure test that follows are on the pressure testing page.